

Survival Analysis of a Randomised Cancer Trial

Does adjuvant chemotherapy delay recurrence and prolong survival in colon cancer?

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1 Background and question

A central job of data science in health is to answer one question rigorously: **does a treatment actually work, and for whom?** The cleanest evidence comes from a randomised controlled trial (RCT), where patients are assigned to treatments by chance so the groups are comparable.

This report analyses a real, published RCT: the **colon cancer adjuvant-therapy trial** (Moertel and colleagues). After surgery, **929 patients** with stage B or C colon cancer were randomly assigned to one of three arms:

- **Obs**, observation only (the control),
- **Lev**, levamisole,
- **Lev+5FU**, levamisole plus fluorouracil.

Two outcomes were followed over time: **recurrence** of the cancer and **death**. The analytical challenge is that many patients had not had the event by the end of follow-up (their times are *censored*), so ordinary averages do not work. This is what **survival analysis** is built for.

2 The data and randomisation check

Because treatment was randomised, the arms should look similar at baseline. A balance table is the first thing a trial statistician checks.

```
os |>
  select(rx, age, sex, nodes, node4, differ, extent, obstruct, perfor) |>
  tbl_summary(by = rx,
    label = list(age ~ "Age (years)", nodes ~ "Positive nodes",
      node4 ~ "Nodal burden", differ ~ "Differentiation",
      extent ~ "Local extent", obstruct ~ "Obstruction",
      perfor ~ "Perforation", sex ~ "Sex")) |>
  add_p() |>
  modify_header(label ~ "**Characteristic**") |>
  modify_caption("**Baseline characteristics by treatment arm**") |>
  as_kable()
```

Table 1: **Baseline characteristics by treatment arm**

Characteristic	Obs N = 315	Lev N = 310	Lev+5FU N = 304	p-value
Age (years)	60 (53, 68)	61 (53, 69)	62 (52, 70)	0.8
Sex				0.028
Female	149 (47%)	133 (43%)	163 (54%)	
Male	166 (53%)	177 (57%)	141 (46%)	
Positive nodes	2 (1, 5)	2 (1, 5)	2 (1, 4)	0.6
Unknown	3	6	9	
Nodal burden				0.7
1-3 nodes	228 (72%)	221 (71%)	225 (74%)	
>4 nodes	87 (28%)	89 (29%)	79 (26%)	
Differentiation				0.5
Moderate	229 (74%)	219 (73%)	215 (72%)	
Poor	52 (17%)	44 (15%)	54 (18%)	
Well	27 (8.8%)	37 (12%)	29 (9.7%)	
Unknown	7	10	6	
Local extent				0.3
Contiguous	20 (6.3%)	12 (3.9%)	11 (3.6%)	
Muscle	38 (12%)	36 (12%)	32 (11%)	
Serosa	249 (79%)	259 (84%)	251 (83%)	
Submucosa	8 (2.5%)	3 (1.0%)	10 (3.3%)	
Obstruction	63 (20%)	63 (20%)	54 (18%)	0.7
Perforation	9 (2.9%)	10 (3.2%)	8 (2.6%)	>0.9

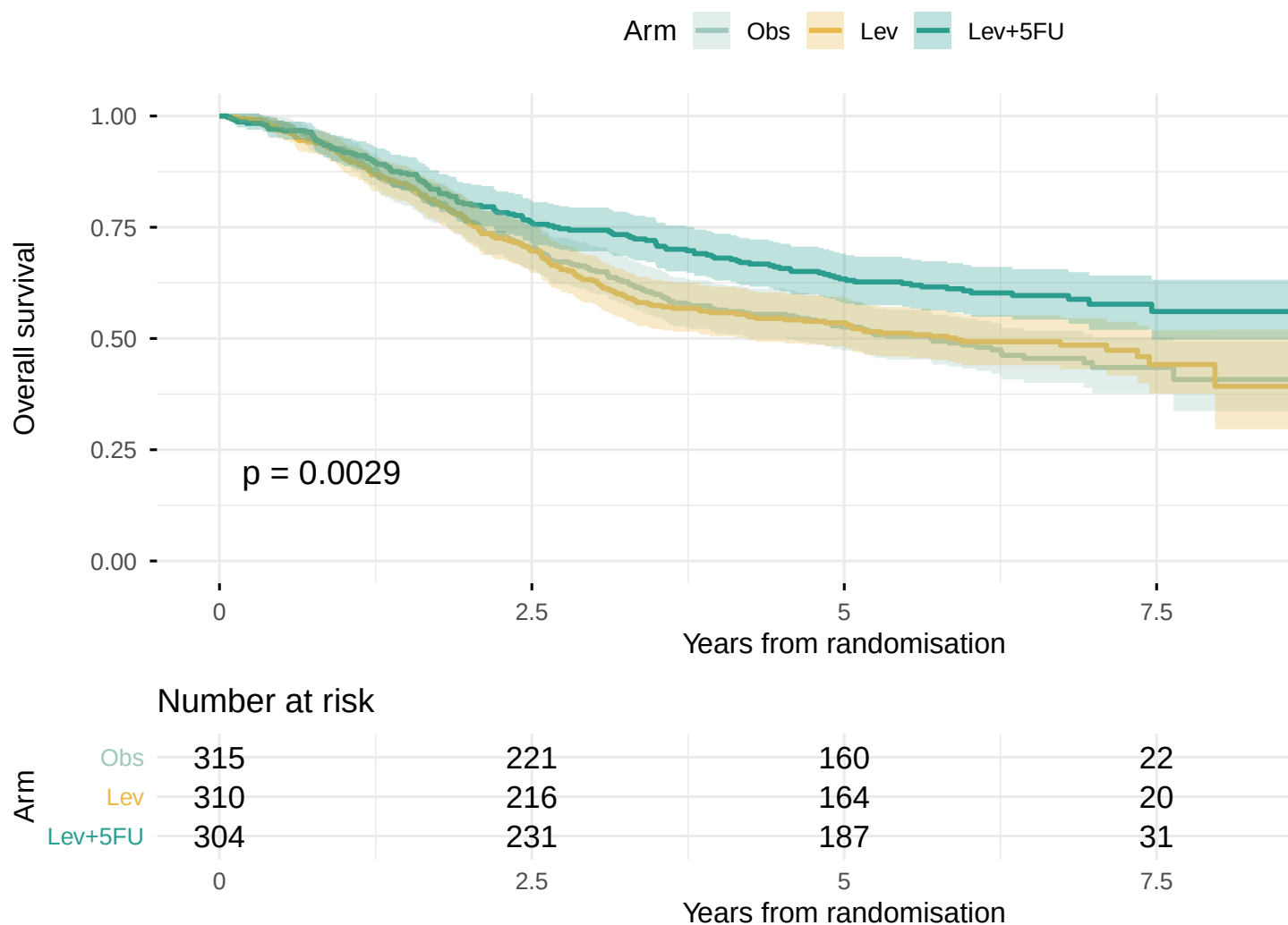
The arms are well balanced: no baseline characteristic differs meaningfully across them (all p-values are large). That is exactly what randomisation should deliver, and it means any later difference in outcomes can be attributed to treatment rather than to sicker patients landing in one arm.

3 Kaplan-Meier survival curves

The Kaplan-Meier estimator turns censored follow-up times into a survival curve: the share of patients still event-free as time passes. We compare arms with the **log-rank test**.

3.1 Overall survival (time to death)

```
fit_os <- survfit(Surv(time_yrs, status) ~ rx, data = os)
ggsurvplot(fit_os, data = os, palette = PAL, conf.int = TRUE,
  risk.table = TRUE, pval = TRUE, censor = FALSE,
  xlab = "Years from randomisation", ylab = "Overall survival",
  legend.title = "Arm", legend.labs = c("Obs", "Lev", "Lev+5FU"),
  risk.table.height = 0.28, ggtheme = theme_minimal(base_size = 12))
```



```
lr_os <- survdiff(Surv(time_yrs, status) ~ rx, data = os)
p_os <- 1 - pchisq(lr_os$chisq, df = length(lr_os$n) - 1)
med <- summary(fit_os)$table[, "median"]
cat("Log-rank p-value (overall survival):", signif(p_os, 3), "\n")
```

```
## Log-rank p-value (overall survival): 0.0029
```

```
cat("Median survival not reached where shown as NA (over half still alive).\n")
```

```
## Median survival not reached where shown as NA (over half still alive).
```

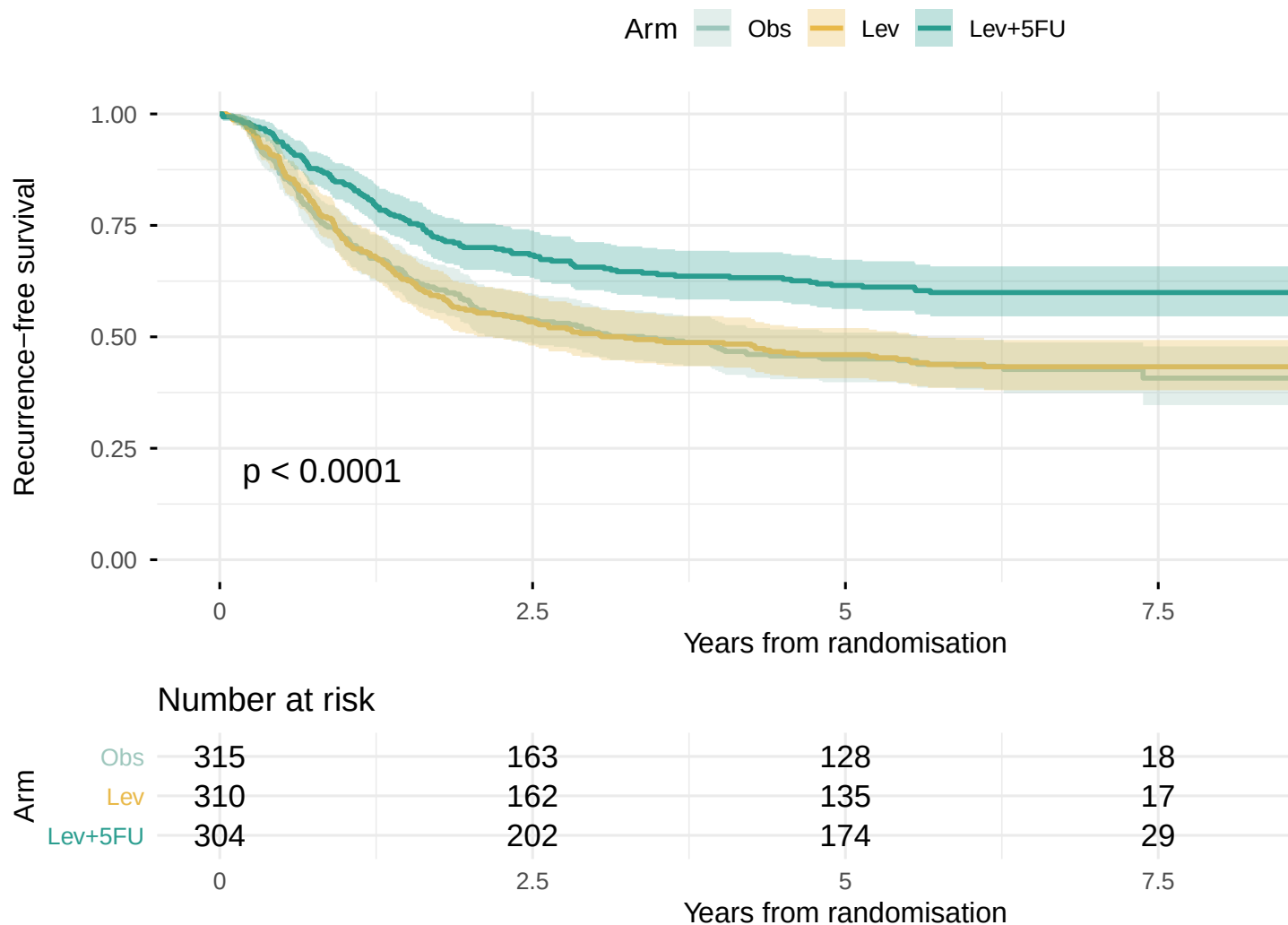
```
print(round(med, 2))
```

```
##      rx=Obs      rx=Lev rx=Lev+5FU  
##      5.70      5.89      NA
```

The **Lev+5FU** curve sits clearly above the other two, and the log-rank test confirms the arms differ significantly. More than half of the Lev+5FU patients were still alive at the end of follow-up, so their median survival is not even reached. Levamisole alone tracks the control arm closely.

3.2 Recurrence-free survival (time to recurrence)

```
fit_rfs <- survfit(Surv(time_yrs, status) ~ rx, data = rfs)  
ggsurvplot(fit_rfs, data = rfs, palette = PAL, conf.int = TRUE,  
            risk.table = TRUE, pval = TRUE, censor = FALSE,  
            xlab = "Years from randomisation", ylab = "Recurrence-free survival",  
            legend.title = "Arm", legend.labs = c("Obs", "Lev", "Lev+5FU"),  
            risk.table.height = 0.28, ggtheme = theme_minimal(base_size = 12))
```



The same story appears earlier and even more strongly for recurrence: the combination arm keeps far more patients cancer-free, and most of its benefit is in preventing the cancer from coming back in the first place.

4 Cox proportional-hazards model

The Kaplan-Meier curves compare arms on their own. The **Cox model** estimates the effect of treatment while adjusting for prognostic factors, and reports a **hazard ratio (HR)**: the relative rate of death. HR below 1 means lower risk.

```
cox <- coxph(Surv(time_yrs, status) ~ rx + age + sex + nodes + extent +
             differ + obstruct + perfor, data = os)
tbl_regression(cox, exponentiate = TRUE,
               label = list(rx ~ "Treatment arm", age ~ "Age",
                             sex ~ "Sex", nodes ~ "Positive nodes",
                             extent ~ "Local extent", differ ~ "Differentiation",
                             obstruct ~ "Obstruction", perfor ~ "Perforation")) |>
  modify_caption("**Adjusted hazard ratios for death (Cox model)**") |>
  as_kable()
```

Table 2: Adjusted hazard ratios for death (Cox model)

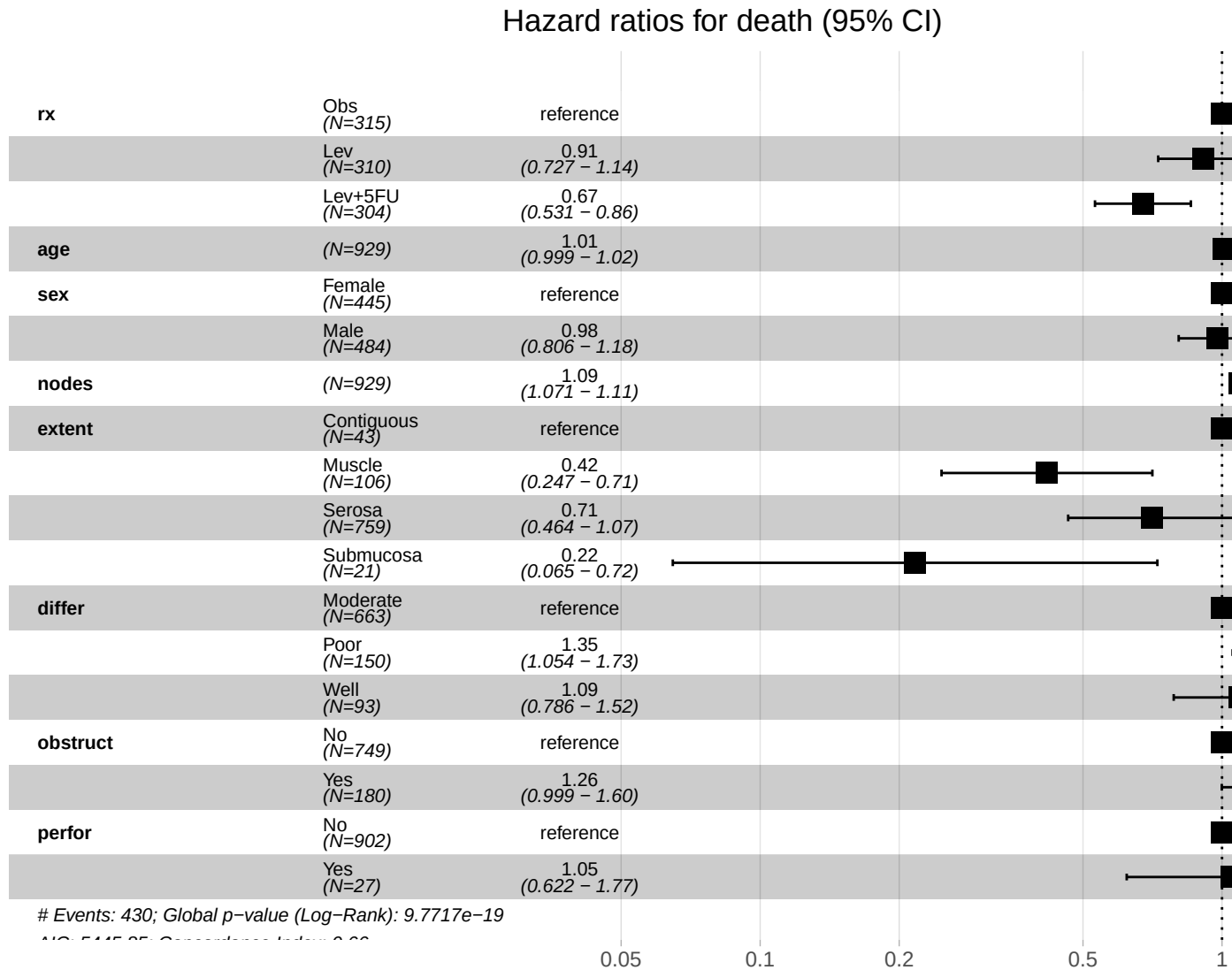
Characteristic	HR	95% CI	p-value
Treatment arm			
Obs	<U+2014>	<U+2014>	
Lev	0.91	0.73, 1.14	0.4
Lev+5FU	0.67	0.53, 0.86	0.001
Age	1.01	1.00, 1.02	0.090
Sex			
Female	<U+2014>	<U+2014>	
Male	0.98	0.81, 1.18	0.8
Positive nodes	1.09	1.07, 1.11	<0.001
Local extent			
Contiguous	<U+2014>	<U+2014>	
Muscle	0.42	0.25, 0.71	0.001
Serosa	0.71	0.46, 1.07	0.10
Submucosa	0.22	0.06, 0.72	0.013
Differentiation			
Moderate	<U+2014>	<U+2014>	
Poor	1.35	1.05, 1.73	0.018
Well	1.09	0.79, 1.52	0.6
Obstruction			
No	<U+2014>	<U+2014>	
Yes	1.26	1.00, 1.60	0.051
Perforation			
No	<U+2014>	<U+2014>	
Yes	1.05	0.62, 1.77	0.9

```
hr <- tidy(cox, exponentiate = TRUE, conf.int = TRUE) |>
  filter(term == "rxLev+5FU")
cat(sprintf("Lev+5FU vs Obs: HR = %.2f (95%% CI %.2f to %.2f), p = %.4f\n",
  hr$estimate, hr$conf.low, hr$conf.high, hr$p.value))
```

```
## Lev+5FU vs Obs: HR = 0.67 (95% CI 0.53 to 0.86), p = 0.0012
```

After adjusting for age, nodal burden, tumour extent and differentiation, the combination arm reduces the rate of death by roughly **one third** compared with observation, and the effect is statistically significant. The number of positive lymph nodes and the local extent of the tumour are the strongest prognostic drivers, as clinical knowledge would predict.

```
ggforest(cox, data = os, main = "Hazard ratios for death (95% CI)")
```



5 Checking the proportional-hazards assumption

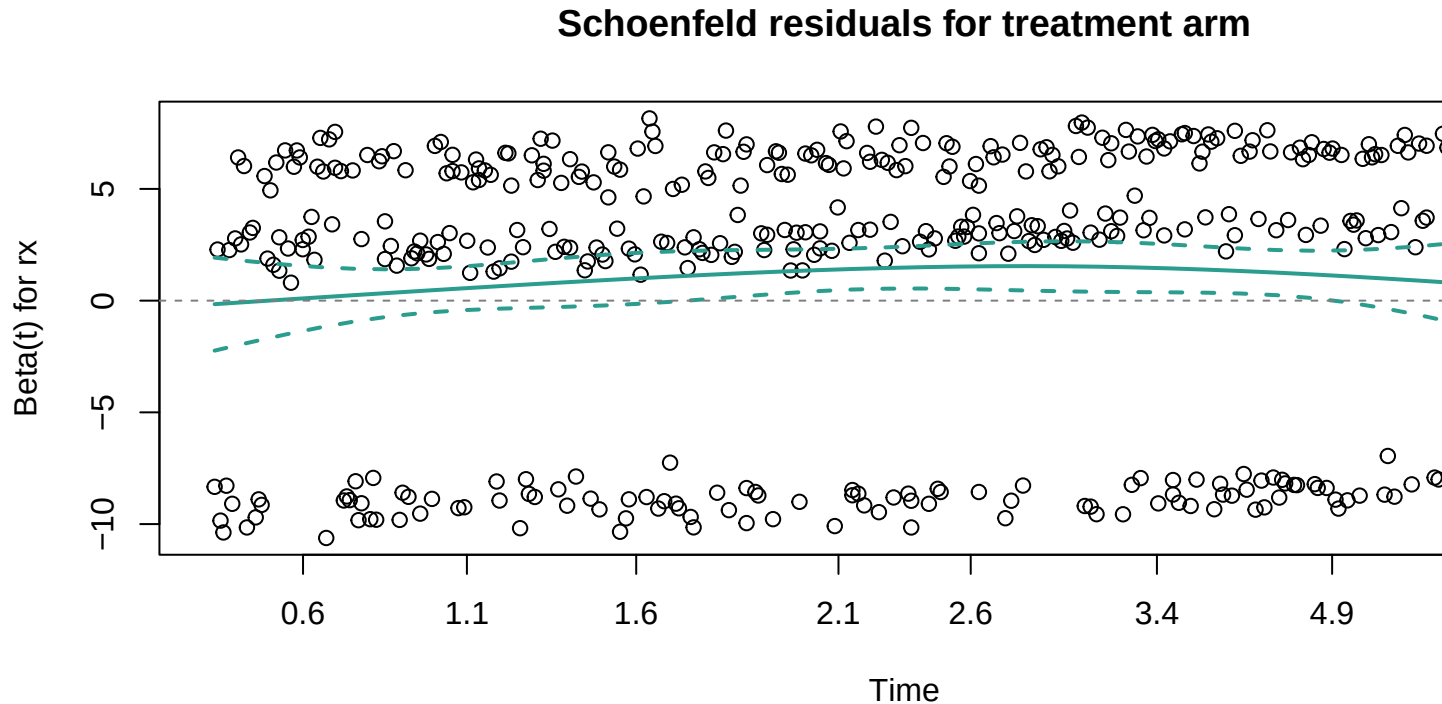
The Cox model assumes a treatment's effect on the hazard is roughly constant over time. We test this with scaled Schoenfeld residuals.

```
zph <- cox.zph(cox)
print(zph)
```

```
##          chisq df      p
## rx          2.9209  2 0.23213
## age         1.1807  1 0.27722
## sex         0.3537  1 0.55202
## nodes       0.0203  1 0.88662
## extent      7.1654  3 0.06681
## differ     16.8839  2 0.00022
```

```
## obstruct 6.8003 1 0.00911
## perfor 0.5598 1 0.45434
## GLOBAL 37.9215 12 0.00016
```

```
plot(zph[1], col = ACC, lwd = 2)
abline(h = 0, lty = 2, col = "grey50")
title("Schoenfeld residuals for treatment arm")
```



The global test and the treatment term are not significant, so the proportional-hazards assumption is reasonable for the treatment effect: the hazard ratios above are a fair summary across the follow-up period.

6 A machine-learning benchmark: Random Survival Forest

Cox is the clinical standard, but it assumes effects are linear and additive. A **Random Survival Forest (RSF)** is a machine-learning model that captures non-linearities and interactions automatically. We compare the two on discrimination, measured by the concordance index (C-index, higher is better), and read off which factors the forest finds most important.

```
set.seed(42)
rsf_dat <- os |>
  select(time_yrs, status, rx, age, sex, nodes, extent, differ,
         obstruct, perfor) |> na.omit()
rsf <- rfsrc(Surv(time_yrs, status) ~ ., data = rsf_dat,
            ntree = 600, importance = TRUE)
```



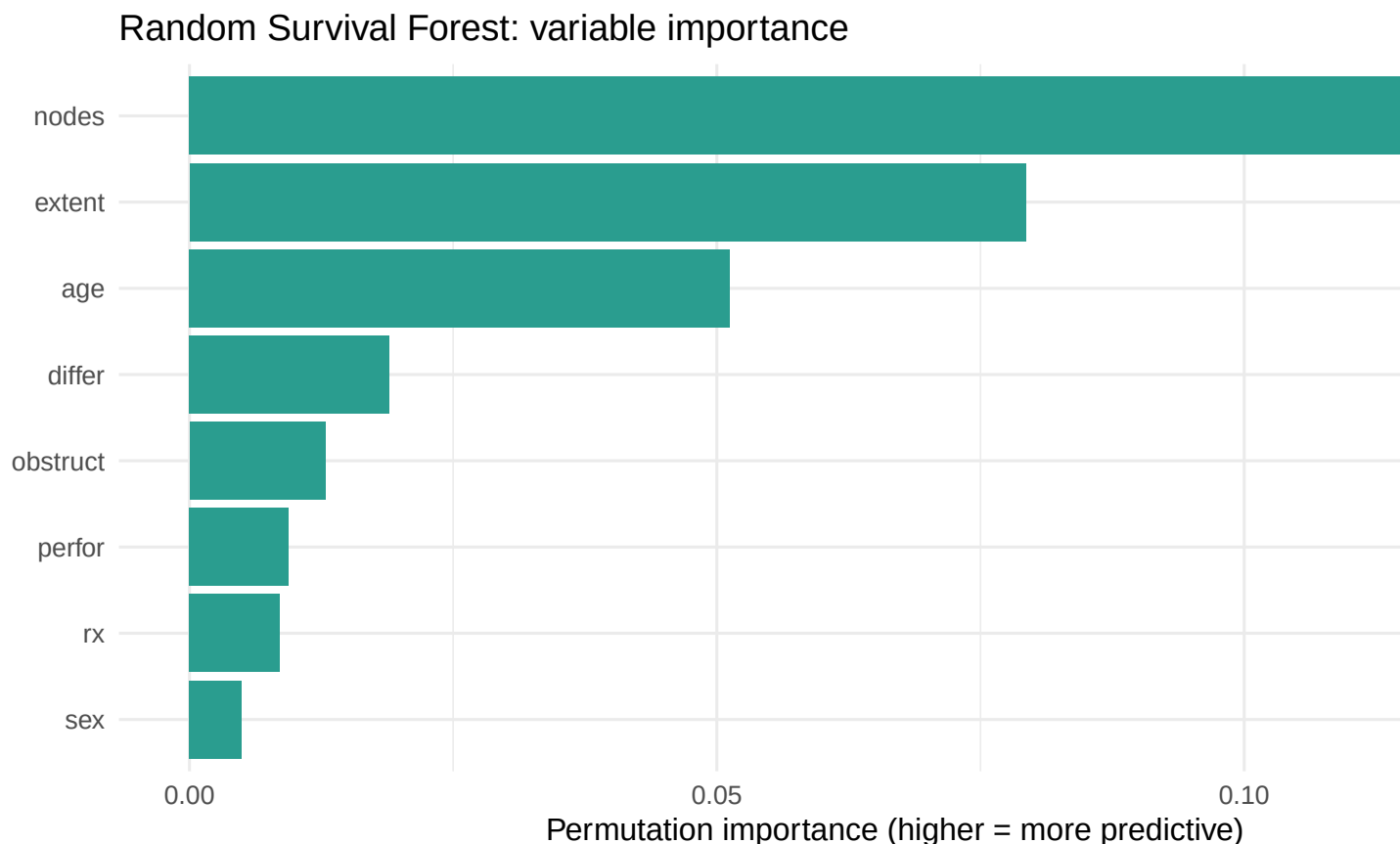
```
c_rsf <- 1 - rsf$err.rate[rsf$ntree]
c_cox <- summary(cox)$concordance[1]
cat(sprintf("C-index, Cox model           : %.3f\n", c_cox))
```

```
## C-index, Cox model           : 0.663
```

```
cat(sprintf("C-index, Random Survival Forest: %.3f\n", c_rsf))
```

```
## C-index, Random Survival Forest: 0.621
```

```
vi <- sort(rsf$importance, decreasing = TRUE)
vidf <- data.frame(var = names(vi), imp = as.numeric(vi))
ggplot(vidf, aes(reorder(var, imp), imp)) +
  geom_col(fill = "teal") + coord_flip() +
  labs(title = "Random Survival Forest: variable importance",
       x = NULL, y = "Permutation importance (higher = more predictive)")
```



The two models reach **very similar discrimination**, which is reassuring: the straightforward Cox model is not leaving much on the table here. The forest ranks the same drivers at the top (nodal burden, tumour extent, age), and it also picks up treatment arm as informative. When a simple, interpretable model matches a flexible one, the simple model is usually the right choice to report.

7 Interpretation and clinical recommendation

- **The combination works.** Adjuvant **levamisole plus fluorouracil** clearly delays recurrence and prolongs survival in stage B/C colon cancer, cutting the rate of death by about **33%** versus observation after adjustment, consistent across the Kaplan-Meier curves, the log-rank tests and the Cox model.
- **Levamisole alone does not.** Its survival curve sits on top of the control arm; the active ingredient is the fluorouracil combination.
- **Most of the benefit is in preventing recurrence**, which is where the curves separate earliest and widest.
- **Target the highest-risk patients.** Patients with more than four positive nodes and greater local extent have the worst prognosis and the most to gain, so they are the clearest candidates for the combination regimen.

8 Limitations

This is a single historical trial, so the specific drug regimen reflects the standard of its era rather than today's protocols; the value here is the method, not a current treatment recommendation. Some covariates have missing values (handled by complete-case analysis for the models). Survival analysis estimates associations adjusted for measured factors only; unmeasured confounding is controlled by randomisation for the treatment comparison but not for the observational prognostic factors.

Built in R (survival, survminer, randomForestSRC, gtsummary) from the colon cancer adjuvant-therapy trial, a real published RCT with openly available patient-level data. Author: Kingsley Amegah.